



Trust and reliance on AI — An experimental study on the extent and costs of overreliance on AI

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ABSTRACT

Decision-making is undergoing rapid changes due to the introduction of artificial intelligence (AI), as AI recommender systems can help mitigate human flaws and increase decision accuracy and efficiency. However, AI can also commit errors or suffer from algorithmic bias. Hence, blind trust in technologies carries risks, as users may follow detrimental advice resulting in undesired consequences. Building upon research on algorithm appreciation and trust in AI, the current study investigates whether users who receive AI advice in an uncertain situation overrely on this advice — to their own detriment and that of other parties. In a domain-independent, incentivized, and interactive behavioral experiment, we find that the mere knowledge of advice being generated by an AI causes people to overrely on it, that is, to follow AI advice even when it contradicts available contextual information as well as their own assessment. Frequently, this overreliance leads not only to inefficient outcomes for the advisee, but also to undesired effects regarding third parties. The results call into question how AI is being used in assisted decision making, emphasizing the importance of AI literacy and effective trust calibration for productive deployment of such systems.

1. Introduction

While technical innovations in artificial intelligence (AI) are evolving rapidly, one practical area that AI has already heavily impacted is decision-making. Where humans previously had to rely on their own expertise, nowadays complex decisions in many domains are supported by AI systems, such as hiring (Li et al., 2021), investment advice (Lourenço et al., 2020), loan approval (Sheikh et al., 2020) and justice (Re & Solow-Niederman, 2019). Benefits of adopting AI can include more informed decisions, as the AI can process significant amounts of information effectively, or fairer results due to the removal of human flaws such as cognitive bias (Agarwal et al., 2018; Bozdog, 2013; Savulescu & Maslen, 2015). However, sometimes AI recommendations can be erroneous and misleading. Factual errors or algorithmic bias can lead to unethical treatment of third parties,¹ unprofitable outcomes for

businesses,² or even safety hazards when operators fail to take control in a situation where a machine errs.³ Therefore, blindly (over-)relying on these technologies carries risks and if trust in AI is too high, users may fail to recognize mistakes and follow advice that could be harmful to others and to themselves.

The current study investigates whether users overrely on AI advice — to their own detriment and potentially that of other parties. We conduct an incentivized, behavioral experiment based on an adapted trust game that was complemented with AI-generated advice. We analyze whether participants overrely on it, that is, whether they follow the AI advice even when it contradicts available context information as well as their own assessment, resulting in inefficient outcomes.

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¹ See e.g., the infamous case of a recruitment AI that turned out to be biased against women, <https://incidentdatabase.ai/cite/37/> (last accessed June 06, 2024).

² See e.g., an inaccurate predictive pricing algorithm contributing to the shutdown of a real estate division, <https://incidentdatabase.ai/cite/149/> (last accessed June 06, 2024).

³ See e.g., a report of a vehicle colliding when a “driver trusted the system [...] and failed to intervene in time”, <https://incidentdatabase.ai/cite/426/> (last accessed June 06, 2024).

2. Related literature

2.1. Algorithm aversion versus appreciation

Whether overreliance on AI advice is a risk worth taking seriously may be called into question when one refers to the wide selection of literature on *algorithm aversion*. Dietvorst et al. (2015) coined the term to describe the tendency of people to avoid using algorithms after observing their errors, even when these algorithms outperform human advisors. This concept aligns with broader research on trust in automated systems, which indicates that trust is dynamic and can be significantly eroded by mistakes. Users often require a prolonged period of error-free interactions to regain trust in a system that has previously failed (Hoff & Bashir, 2015; Kraus et al., 2019; Lee & See, 2004). Numerous studies have since demonstrated that individuals frequently avoid relying on decision-making systems in various scenarios (e.g. Bigman & Gray, 2018; Yeomans et al., 2019).

However, it has also been shown that in many situations, people do prefer algorithms, making overreliance a sensible worry. Logg et al. (2019) identified a contrasting behavior known as *algorithm appreciation*, where people tend to prefer algorithmic judgment over human judgment, even when the algorithm's processes are not fully transparent.⁴ This perspective is supported by subsequent research, which has underscored the potential risk of overreliance on AI (cf. Araujo et al., 2020; Candrian & Scherer, 2022). Several scholars have attempted to reconcile potential differences between algorithm aversion and appreciation, partially explaining them with different preferences depending on judgment domain and context. Whether algorithm aversion or algorithm appreciation emerge depends on several factors. These include the nature of the task (objective versus subjective) (Castelo et al., 2019; Thurman et al., 2019); characteristics of the algorithm or the human agent, such as perceived performance and expertise (Jussupow et al., 2020); the specific framing of the study such as the description of the human and the algorithmic agent (Hou & Jung, 2021); and variations in increased benefits and decreased user control associated with algorithms (Schaap et al., 2023). Lately, Morewedge (2022) proposed a theoretical framework arguing that algorithm aversion originates from biased human self-evaluations, which are highly domain-specific as they are modulated by the identity-relevance of the judgment and the ambiguity of its evaluative criteria, further arguing that algorithm aversion is sometimes mistakenly understood as domain-general.

2.2. (Over-)trust in AI

Despite their differences, the above-mentioned widely-cited studies by Dietvorst et al. (2015) and Logg et al. (2019), as well as most recent research supporting either algorithm aversion or appreciation, have in common that the influence of trust is barely mentioned or analyzed. However, trust is a major predictor of whether people will actually rely on AI advice (Kohn et al., 2021) and various models from the trust in automation literature include the connection between trust and reliance (cf. Hoff & Bashir, 2015; Lee & See, 2004; Schaefer et al., 2016). In our study, we utilize the definition of Mayer et al. (1995) defining trust as “the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular

action important to the trustor” (Mayer et al., 1995, p. 712).⁵ The definition allows us to differentiate between the willingness to assume risk, equating trust, and the action of taking risk, which in turn corresponds to trusting behavior (Mayer et al., 1995). In the context of human-computer interactions, one pivotal definition of trust continues this distinction by separating trust as an attitude and reliance as the associated behavior (Lee & See, 2004). Adopting this terminology, our interest lies not only in observing reliance as the behavioral outcome of trust, but also in analyzing attitudinal measures of the underlying psychological trust construct.

Conventional automated decisions-support systems, including those studied by Dietvorst et al. (2015) and Logg et al. (2019), were specialized on singular solutions and were mostly perceived as supportive tools (cf. De Visser et al., 2018). However, the rise of Large Language Models (LLMs) such as GPT-4 and the subsequent public exposure to conversational AI have changed the dynamics of human-computer interaction, gradually transforming machines from tools to partners and revitalizing the importance of the long-standing research on trust (Glikson & Woolley, 2020). When used correctly, LLMs and AI in general have considerable potential to improve decision-making. However, to foster successful utilization of autonomous systems such as AI advisors, appropriate trust is vital, as a lack of trust might result in misuse, leading to lower rates of adoption or outright rejection of beneficial technology. Yet, excessive trust, known as *overtrust*, can lead people to misuse technology, blindly relying on systems that do not warrant that trust. Such overtrust, defined as poorly calibrated trust exceeding system capabilities (Lee & See, 2004), can lead to undesired consequences both from an economic as well as an ethical perspective (Parasuraman & Riley, 1997).

Extant research provides inconsistent evidence regarding the question whether people avoid or overrely on AI advice and how this behavior is related to distrust or overtrust in AI. As various scholars argue that observed differences stem from characteristics that differ between domains, one way to address this issue lies in domain-independent study designs as elaborated in the following chapter.

3. Study design and predictions

So far most prevalent studies rely heavily on complex context descriptions and are highly domain-specific, e.g. examining advice in medical (Longoni et al., 2019; Prah & Van Swol, 2017), legal (Araujo et al., 2020; Bigman & Gray, 2018) and dating or financial domains (Castelo et al., 2019; Schaap et al., 2023), and there has been a lack of domain-agnostic studies aiming to complement the literature with a more generalizable approach. Although some prior studies have included rather abstract tasks, such as guessing the weight of people or forecasting the popularity of songs (Logg et al., 2019), such tasks entail

⁴ A common misunderstanding in the literature is that Logg et al. (2019) directly oppose Dietvorst et al. (2015). However, Logg et al. clarify in their own paper that their findings do not contradict Dietvorst et al.'s; While Logg et al. examined appreciation for algorithms before any mistakes occurred, Dietvorst et al. identified that people avoid algorithms after witnessing their mistakes.

⁵ This definition has been created in the context of human interactions, but studies showed that there are differences between interpersonal trust and trust in machines. Thus, we acknowledge that in some contexts other definitions might be more appropriate for human-machine trust. However, for our purposes, this definition is applicable, as many elements that distinguish human-human interactions from human-machine interactions are not present in our experiment. For example, in our online setting, there is a one-shot interaction with an AI advisor, thus there is no opportunity for trust to develop dynamically, be broken or be mended as is the case in prolonged interactions. Instead, we are focusing on initial trust in an advisor and the associated decision whether to rely on the advice, and are presenting the interaction in an almost sterile context. One essential aspect of the definition in regard to our study is the willingness to be vulnerable, as we examine whether people will decide to follow advice, thus accepting vulnerability by risking their payoffs. This is the case for both, human and AI-generated advice, that is virtually undistinguishable from human recommendations. In line with prior research, we therefore apply this definition to trust in AI, as it “does not limit the concept of trust to human-human interaction” (Glikson & Woolley, 2020, p. 630).

low impact on the user and barely resemble real cases of AI-assisted decision making (Schaap et al., 2023). In contrast, our experiment aims to be as general as possible while retaining foundational properties of most contemporary AI advice use cases: A setting of ambiguity, advice on an uncertain outcome, potential benefit, risk and human interaction. A majority of AI-assisted decisions take place in risky environments and impact human interaction one way or another, whether it is medical advice affecting the doctor-patient interaction, human resources advice affecting the recruiter-applicant interaction or legal advice affecting the judge-defendant interaction. To mirror this relevance, we are examining how participants react to advice generated by an AI, in an interactive setting that includes a third party.

To achieve this, we are using a modified trust game, in which two players have to decide whether to cooperate with each other. One player, the "proposer," receives an initial endowment and must decide whether to send it to another player, the "responder." If they send the money, the amount is quadrupled. The responder then decides whether to keep the entire sum, or to send half of it back to the proposer. That is, the proposer can double his endowment when sending the money, but they also risk losing everything if the responder refuses to send a portion back. The players play three rounds without any decision aid. In the fourth round, however, some participants in the role of player A receive advice on whether to cooperate with the other player. This advice is either labelled as 'expert advice' or 'AI advice,' which allows us to cleanly separate the effect of receiving advice from an AI vs. the effect of receiving advice per se.

Therefore, our task is widely domain-independent, while still encompassing core elements of AI assisted decision making, cooperation between human actors, and a financially incentivized motivation for the participants. This setup is conceptually simple, involves taking a binary decision and has been widely used in experimental economics to study human interactions in general. By adding a generative AI as an advisor, we can measure its effects on participants' reliance without requiring elaborate context descriptions, domain specifics or prior knowledge. One of our goals is therefore to shed light on how people react to AI advice in a more general setting, without being influenced by specific contexts or domains that often may have caused algorithm aversion or appreciation in the first place (Dietvorst et al., 2015; Logg et al., 2019).

This methodological choice further allows us to distinguish between participants' trust attitude towards the AI and their reliance on it. In extant empirical research, trust and reliance often have not been sufficiently differentiated conceptually and various studies have only measured one of the two constructs (Kohn et al., 2021; Scharowski et al., 2022). Many studies only assess trust via questionnaires, but this offers little evidence for inferences regarding actual behavior, which is of vital importance for practical implications. However, solely focusing on behavioral measures of reliance alone, as seen in most algorithm aversion/appreciation studies, does not permit deductions regarding underlying trust (Körber, 2019; Lee & See, 2004; Miller et al., 2016). In this regard we answer the call for distinguishing and measuring both concepts (Scharowski et al., 2022). One of the advantages of our controlled experimental set-up is, that there are no barriers between trusting (willingness to assume risk) and relying (actually taking risk). Simultaneously, there is no mechanism to force participants to rely on the AI, placing the decision solely into their hands. Consequently, in accordance with prior studies (Kohn et al., 2021; Pearson et al., 2019), we are predicting that participants with higher trust in the advisor will also exhibit higher reliance on advice. Still, our interest regarding overtrust in AI, lies primarily on the behavioral consequences of said overtrust, namely the question whether people will fail to reject misleading AI advice, i.e. overrely on it.

Overreliance on AI advice has been shown to lead to lower cognitive engagement and impaired task performance in prior studies (Alexander et al., 2018). However, a major part of the extant empirical literature was either survey-based, where typically no stakes are involved and responders may be subject to self-report bias, or conducted behavioral

experiments which were limited by focusing solely on the effects on the advisee (for reviews see Glikson & Woolley, 2020; Hou & Jung, 2021; Jussupow et al., 2020). However, in many real use-cases such as loan approval or applicant screening, AI advice will affect more people than the advisee alone. Thus, it is important to study how people actually rely on AI advice that could have both, detrimental effects for themselves as well as consequences for other parties. For this purpose, our incentivized behavioral experiment complements prior results by introducing financial stakes for two interacting participants, where every participant's decision may affect the other party. Thus, to earn a higher bonus payment, all participants are motivated to choose deliberately. While participants in the control group make this decision independently, participants in the treatment groups may decide to follow advice regarding this decision. At the same time, they are incurring the risk of losing their bonus if they follow advice that is misleading in that it fails to accurately predict the other player's behavior. This mirrors various real-world scenarios, where AI provides advice in an ambiguous situation, where the outcome is uncertain as it depends on future human interactions, such as AI recommendations on loan approval, where advice may prove to be wrong, if the debtor turns out to default on his obligations. In line with prior literature (Berg et al., 1995; Parasuraman & Riley, 1997), a consequence of our design is that overreliance on AI advice may lead to undesired pay-off changes for both the advisee (the 'proposer') and a third party (the 'responder'). Consequently, players are incentivized not to blindly trust unfitting advice – and yet, we expect that participants will trust AI regardless, exhibiting overreliance, even if the advice clashes with available information, and detrimentally affects outcomes.

To test this proposition, we are splitting our research question into three testable predictions.

1. Decision-makers are likely to follow AI advice—and to a greater extent than advice per se.
2. Decision-makers are likely to overrely on AI advice in that they follow it even when said advice contradicts available context information.
3. Overreliance is costly in that it leads to inefficient outcomes for both the advisee and the interacting partner.

In summary, our study aims to offer the following contributions: It a) enhances the algorithm aversion and appreciation debate by shedding light on human behavior in a domain-independent setting including a generative AI; b) provides an experimental setup that, following experimental economic methods, is incentivized, interactive and deception-free; c) examines the relationship between trust and reliance by assessing both behavioral and attitudinal measures; and d) provides evidence on how AI can lead to overreliance, i.e., to reliance even in situations where the advice is contrary to available context information and against the advisees' own interests and those of others.

4. Method

4.1. Participants

We recruited our participants from Prolific, an online research participant pool that has been frequently used in experimental studies (Palan & Schitter, 2018). All participants were United Kingdom residents who indicated that English was their primary language. Out of the 402 recruited participants, 329 finished the entire study of which 319 passed all attention checks. The final sample ($n = 319$) ranged in age from 18 to 40 ($M = 30$, $SD = 5.56$) years and consisted of 56% male participants. The highest level of education attained ranged from high school to doctorate degree, with undergraduates being the largest subgroup at 44%. At least 75% of the participants were not students and 59% were full-time employees.

4.2. Task

Participants played six periods of a modified trust game (Berg et al., 1995), a classical game in experimental economics, where two players can cooperate to their mutual advantage but where cooperation often fails due to the risk of being exploited by the other player. Participants were matched in groups of two and were randomly assigned the role of the proposer (player A) or the responder (player B). The trust game was played over six periods, and while participant roles remained constant, pairs were rematched after every period in a stranger matching protocol to avoid reputation building (Andreoni & Croson, 2008).

Fig. 1 depicts the payoff structure for players A and B. Each period, both players start with an endowment of 10 ECU. The task of both players is to choose whether to cooperate with each other by choosing 'IN' or 'OUT' (strategy method for the responder B). For player A, the first mover, opting OUT means that A and B keep their initial endowment of 10 ECU and the period ends. Opting IN in contrast, allows for a potential payoff of 20 ECU for A and B if (and only if) player B, the second mover, opts IN as well. If, however, player A chooses IN and player B chooses OUT, player B receives 40 ECU while player A ends the period with nothing. This means that if player A chooses to cooperate with player B, they may either have both their credits doubled in return or have their own credits stolen. Alternatively, they may choose to keep their endowment safe by not opting IN, leading to both players keeping their original endowment. After every period the participants are informed about the choices of A and B and their resulting payoff. This simple design mirrors various real situations where a party can cooperate with another to their mutual benefit, but the decision to cooperate is risky, as the other party's future behavior is unknown. Hence, the setting is appropriate to introduce AI advice, as it adequately captures situations where a recommender AI augments the decision of a human agent by advising a particular course of action in a situation of risk.

Out of the six played periods, our analysis will focus on the fourth period, in which player A receives information about their newly matched player B, in the form of player B's decision history from periods 1–3. That is, in the fourth period, each player A learns which decisions their partner had *actually* taken in the three previous rounds. Knowing B's previous choices helps player A to make an informed guess about player B's behavior in the fourth period, although it remains impossible, to reliably predict player B's decision, because B will make a choice in real time, only after player A has received this information and made their choice.

4.3. Design

The study employed a $2 \times 2 + 1$ between-subjects design, with the control group maintaining an unchanged experimental setup as explained above, without any advice. Manipulations involved a third-

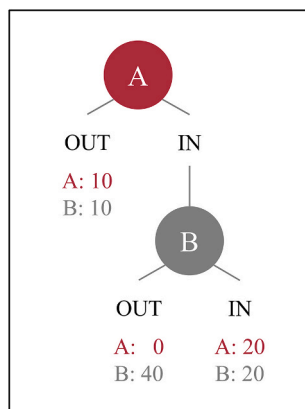


Fig. 1. Payoff structure for players A and B in the utilized trust game.

party advisor who provides advice to player A in period 4, where source (AI vs. Expert) and content (cooperation encouraged vs. discouraged) were manipulated. Depending on the treatment, player A received either AI advice encouraging (AI +) or discouraging (AI -) cooperation with player B, or they received expert advice encouraging (Ex +) or discouraging (Ex -) cooperation. Participants were randomly assigned to one of the five treatment groups, resulting in the following distribution: 69 (AI +), 66 (AI -), 58 (Ex +), 61 (Ex -) and 65 (control).

Depending on the treatment participants were informed that they will receive a recommendation by an AI or by an expert, with no indication that the expert might be an AI in the latter case. A manipulation check asking what kind of advice the participants received, provided evidence that this manipulation worked as intended, as participants in the expert treatment chose the answer “expert advice,” and not “AI advice” (and vice versa). Further, the participants were informed that the AI/expert has knowledge of the trust game and player B's decision history when creating the recommendation. Therefore, while it is possible that individual participants would expect the AI to be more accurate than the expert, there was no reason to believe that in generating its recommendation, the AI had more information about player B than the expert.

Substantially, we were interested in the effect that AI advice had relative to no advice (control group). We added the additional expert advice treatment to be able to evaluate whether the effect of AI advice would go beyond the effect of advice per se. If reliance is higher in the AI treatments than in the expert treatments, we conclude that participants rely on the advice *because they knew it came from an AI*. To this aim, the wording of advice was identical between AI and Expert treatments, ensuring the difference between the AI and the expert treatments was that participants were aware of the advice being AI-generated. As the advice player A received was based on B's individual history, and there were 8 possible decision histories and two potential recommendations (encouraging or discouraging cooperation), we finally had 16 unique recommendation texts. As is the norm in experimental economics, we did not want to deceive participants. It would have been clearly deceptive to tell participants that the advice was AI-generated when in fact we had written it ourselves. Therefore, we provided advice that was generated by ChatGPT, OpenAI's LLM-based chatbot, that we had prompt-engineered to provide recommendations either encouraging cooperation or non-cooperation for every scenario.⁶ Table 1 provides examples (see Appendix Item 1 for all advice texts).

Participants were aware that the advice could not always be correct, as it was provided before player B made a choice. As player B was a real person making decisions live during the experiment, it was impossible to precisely predict every B's choice. Whether the advice was correct, was only discovered after period 4, when participants would be informed about both players' choices and their resulting payoff. By design, participants could not leverage this learning however, as there was no further interaction with the advisor. This was a deliberate design choice as the study's focus was to examine whether people rely on advice specifically because it is AI-generated, not because of its historical accuracy.

In the following, we will use the term ‘reliance’ to describe participants who follow advice. We will refer to this behavior as ‘overreliance’ if aggregate behavior across treatments satisfies two additional conditions: 1) Participants follow advice even when it contradicts available context information; and 2) reliance leads to inferior outcomes. These conditions reflect important features of our experiment because they allow us to analyze whether participants systematically fail to reject advice in implausible scenarios and what consequences this has. As participants receive either cooperation-encouraging or cooperation-

⁶ Note that these recommendations were generated prior to the experiment to ensure that all participants receive identical advice in the same situation, depending on treatment and history scenario.

Table 1
Two exemplary advice texts for player B's history scenario IN-OUT-IN.

History scenario	Advice encouraging cooperation	Advice discouraging cooperation
IN-OUT-IN	In this situation, I advise you to trust player B. Although player B defected in the second round, they cooperated in the first and third rounds, indicating that they may have a preference for mutual cooperation. By trusting player B, you signal your own trustworthiness and encourage them to reciprocate with cooperation.	In this situation, I advise you not to trust player B, as their defection in the second round suggests that they are not reliable partners for mutual cooperation. Even if they cooperated in the first and third rounds, their earlier defection signals a lack of commitment to reciprocal behavior and a potential desire for exploitation.

discouraging advice for each of the eight possible decision histories, we were able to test whether participants would overrely on AI advice to their own detriment (e.g., following advice encouraging cooperation when facing a player B with an OUT-OUT-OUT history) and that of third parties (e.g., following advice discouraging cooperation when facing a player B with an IN-IN-IN history).

4.4. Procedure

Conducted on the online platform LIONESS (Giamattei et al., 2020), our experiment lasted 15–20 min in total, with each participant earning an average of approx. 3 £. Data collection occurred over a week in May 2023, with each treatment featured in at least four of six sessions to avoid time-fixed effects. Prior to conducting this study, we obtained approval from our university's ethics committee.

Participants, redirected to the LIONESS site upon study acceptance, completed an introductory section, followed by the interactive study and a post-experimental questionnaire. The introduction covered confidentiality, informed consent, and general instructions, as well as a comprehension test. Randomly assigned to one of five treatment groups, participants received specific study instructions. The instructions were identical for all participants except for the control group members who did not receive any advice in period 4 and thus were not informed about it. Afterwards the interactive task commenced as described in section 4.2. When the manipulation occurred in period 4, player A would see player B's history, receive advice depending on the treatment and make a choice. Subsequently, we prompted player A to recall player B's past decisions (periods 1–3) and predict B's choice in period 4 on a 4-point Likert scale ranging from "very likely IN" to "very likely OUT," to further assess the influence of the received advice on the decision.⁷ Control group participants predicted B's choice without advisor information, explaining their reasoning in writing. This belief elicitation occurred immediately after the decision-making process to minimize recall bias. In the meantime, player B answered six general questions regarding risk attitude (Dohmen et al., 2011), to avoid drop out due to forced inactivity. The following periods 5 and 6 were included to mitigate potential last-round effects on treatment outcomes.

Following the completion of all six periods in the initial experimental stage, participants swapped roles and completed another six periods of the trust game. This design enabled a doubling of observations without increasing the subject number. To address potential bias from varying interaction experiences, we conducted separate analyses, distinguishing whether participants encountered the manipulation in stage 1 or stage 2. Results remain qualitatively the same in strength and direction. We therefore pooled all observations.

⁷ While we are aware that this design cannot entirely rule out that the answer could be biased by the advice provided, our research aim prioritized not biasing the behavior.

Finally, we administered a comprehensive set of post-experimental inquiries with 18 questions covering traits independent of the task, such as trust propensity on a 5-item scale (Höddinghaus et al., 2021; Ostendorf & Angleitner, 2004), trust in technology on a 3-item scale (Höddinghaus et al., 2021; Mcknight et al., 2011) and affinity for technology interaction on a 4-item scale (Franke et al., 2019). Additionally, the questionnaire included a commitment check, two attention checks, and a question aiming to verify participants' awareness of the advice source, serving as a manipulation check. Finally, participants had the option to read a debriefing about the experiment's nature before receiving their completion code.

4.5. Dependent variables

To test our predictions, we combine objective and subjective measures. In terms of behavior, we are interested in whether participants follow advice, i.e. in whether they rely on it. To examine that, we used two variables to measure participant A's reliance: Cooperation and advice conformity. Cooperation represents the proportion of participants A choosing 'IN'. Advice conformity indicates the share of participants whose choice is in line with the received advice, opting 'IN' when cooperation was encouraged and 'OUT' when discouraged.⁸ Both cooperation rate and advice conformity are analyzed at an aggregate, not an individual level, i.e., we compare distributions across different groups instead of observing individual choices. This means that while we cannot make statements about which individuals in particular overrelied on the advice, we are able to examine whether overreliance is exhibited on average. In this regard, differences in cooperation rates compared to the control group allow us to distinguish the effect of reliance on advice from the effect of participants' own rationale when presented with B's decision history. Similarly, while advice conformity alone does not explain how many participants were actually convinced by the advice, observing differences in advice conformity between treatments allows us to examine whether participants' reliance on the advice differed between groups.

The attitude trust as a psychological construct was measured subjectively, adapting the three item scale for situational trust from Thielsch et al. (2018), which ranges from 1 (Strongly Disagree) to 7 (Strongly Agree). An example item was "I completely trusted the AI/expert advisor." Scale items were averaged (reliability $\alpha = 0.89$). Participants also provided self-reported assessments related to the task, including user expertise on a 4-item scale (reliability $\alpha = 0.88$) (Tauchert & Mesbah, 2019), and single items for perceived autonomy, preference for advice and mental demand (Bućinca et al., 2021).

5. Results

Overall, our data are largely consistent with our predictions and provide support for the existence of overreliance on AI. We summarize the observations in the following main results:

Result 1. Decision-makers tend to follow AI advice—and to a greater extent than advice per se.

We begin our analysis with data pooled across all eight decision histories and compare the average cooperation rates across treatments. As the decision histories resulted from the actual behavior of players B during the first three periods, we had no control over their distribution. However, as expected due to the random assignment of treatments, an ex post analysis revealed that player B's decision histories were evenly distributed, i.e. there were no significant differences in their frequencies across the five treatments. As depicted in Fig. 2, in all treatments advice affects the cooperation rate in the intended direction compared to the control group. However, compared to the control group only the AI

⁸ The control group lacks a value in advice conformity, as participants did not receive advice, but saw only player B's decision history.

treatments lead to significant change, with small effect for encouraging advice ($\chi^2(1) = 4.6625$, $p = 0.031$, $\phi = 0.19$) and medium effect for discouraging advice ($\chi^2(1) = 12.8252$, $p = 0.000$, $\phi = 0.31$). Both expert treatments on the other hand show no significant effect ($\chi^2(1) = 0.0940$, $p = 0.759$, $\phi = 0.03$ for Ex + treatment; $\chi^2(1) = 1.9186$, $p = 0.166$, $\phi = 0.12$ for Ex – treatment). Directly comparing AI vs. expert advice further supports this result, as there is a group difference of small effect size for advice encouraging cooperation ($\chi^2(1) = 3.2386$, $p = 0.072$, $\phi = 0.16$) and particularly for discouraging advice ($\chi^2(1) = 4.7445$, $p = 0.029$, $\phi = 0.19$). Thus, although participants had no additional information on the reliability or the accuracy of the AI, AI advice did significantly change human behavior whereas mere expert advice did not.

When assessing advice conformity instead of cooperation rate, Logit regressions in Table 2 provide further evidence for the above-mentioned pattern of higher reliance on AI advice, as column (1) reveals AI treatments to result in significantly higher advice conformity independent from advice direction. At the same time, we can identify that advice encouraging cooperation also leads to higher advice conformity than discouraging advice. When including self-reported variables in column (2), we observe a strong relationship between subjectively reported situational trust and objectively measured behavior, providing further validation of the positive relationship between trust and reliance. Additionally, we observe that participants' risk attitude and propensity to trust influence their behavior, both having weak negative effects on advice conformity and thus reliance.

Result 2. Decision-makers overrely on AI advice in that they follow it even when said advice contradicts available context information.

So far, behavior was analyzed across all eight decision histories. However, to test whether reliance on AI would qualify as overreliance, we examined cases where the recommendation was not in line with contextual information on player B's prior decisions. We grouped all eight history scenarios from periods 1–3 into two groups either consisting of primarily cooperating ($n = 210$) or primarily defecting histories ($n = 109$), and assessed the cooperation rate (CR) of the control group. This way, we can infer whether advice was in line with or contradictory to the contextual information provided by the decision histories. We observe that both the expert and the AI treatments significantly affect CR relative to the control group (see Fig. 3a). However, there is a difference between mainly cooperative and mainly non-cooperative histories. In cooperative histories, both discouraging AI advice ($\chi^2(1) = 16.1073$, $p = 0.000$, $\phi = 0.45$) as well as discouraging

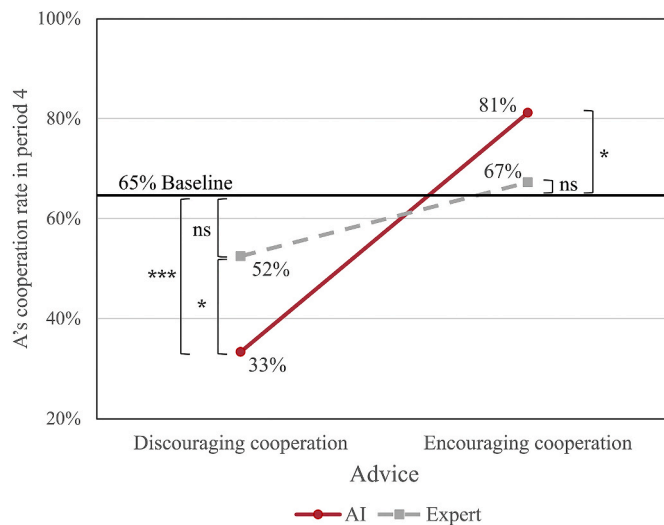


Fig. 2. Player A's mean cooperation rate in period 4 by treatment group, across all history scenarios. Pearson χ^2 p-values.

* $p < .05$; ** $p < .01$; *** $p < .001$.

Table 2

Regression results of advice conformity.

Dependent variable	Advice conformity	Advice conformity
Model	(1)	(2)
<i>Regressors</i>		
AI advice	0.770** (0.275)	1.119** (0.423)
Encouraging advice	0.794** (0.277)	0.999* (0.416)
Situational trust		0.534*** (0.068)
Propensity to trust		-0.102** (0.035)
Risk attitude		-0.057* (0.023)
Constant	-0.088 (0.227)	-2.085* (1.010)
Obs	254	254
Sig	0.001	0.000
Pseudo R ²	0.0510	0.5209

Note: Models present results of a Logit regression. Standard errors are shown in parentheses.

* $p < .05$; ** $p < .01$; *** $p < .001$.

expert advice ($\chi^2(1) = 11.0727$, $p = 0.001$, $\phi = 0.35$) have medium negative effects on CR, and these effects are not significantly different from each other ($\chi^2(1) = 0.8212$, $p = 0.365$, $\phi = 0.1$). For non-cooperative histories however, we observe that encouraging expert advice has no significant effect on CR ($\chi^2(1) = 1.7078$, $p = 0.191$, $\phi = 0.2$), whereas AI advice results in a large effect ($\chi^2(1) = 15.0693$, $p = 0.000$, $\phi = 0.58$) leading to 65% CR compared to 9% in the control group. Given that the difference between AI and expert treatments is highly significant and of medium effect ($\chi^2(1) = 7.5914$, $p = 0.006$, $\phi = 0.42$) although advice texts in both treatments are identical and history scenarios are not statistically different, this delta further supports the proposition that advice affects behavior even stronger when decision makers know it has been provided by an AI rather than an expert advisor.

To verify whether participants actually translated the contextual information provided by decision histories into own beliefs about B's behavior, we asked participants what they believed B would do in period 4. The analysis reveals that participants actually often followed advice dissenting from their personal assessment. That is, participants frequently overrelied on advice, when they knew the recommendation originated from an AI. All players A answered a question asking for their assessment of player B's likely behavior.⁹ We grouped the answers into three groups, "very likely IN" ($n = 96$), "very likely OUT" ($n = 74$) and "ambiguous" ($n = 149$) which encompasses the answers in between. When players A were confident that player B would opt OUT, choosing OUT themselves would be a rational consequence, which was indeed observed in the control group (CR = 6%), as shown in Fig. 3b. Expert advice encouraging cooperation nonetheless, was not significant (ns) in raising CR ($\chi^2(1) = 1.0051$, $p = 0.316$, $\phi = 0.17$). In contrast, AI advice had a significant effect of medium size ($\chi^2(1) = 5.2517$, $p = 0.022$, $\phi = 0.46$). That is, with a CR of 44% almost every second player A decided to risk their endowment by cooperating with a player B who they thought was likely to defect. The same trend was observed in the group judging B to very likely opt IN, where in accordance with their assessment, every A from the control group cooperated with B (CR = 100%). Discouraging advice from an expert did not diminish CR significantly ($\chi^2(1) = 1.3097$, $p = 0.252$, $\phi = 0.18$), whereas AI advice resulted in a significant effect of medium size ($\chi^2(1) = 6.8798$, $p = 0.009$, $\phi = 0.45$). Therefore, every fourth player A chose to forego the potential of a mutually

⁹ There were no significant differences in player A's assessment of player B's behavior between treatments.

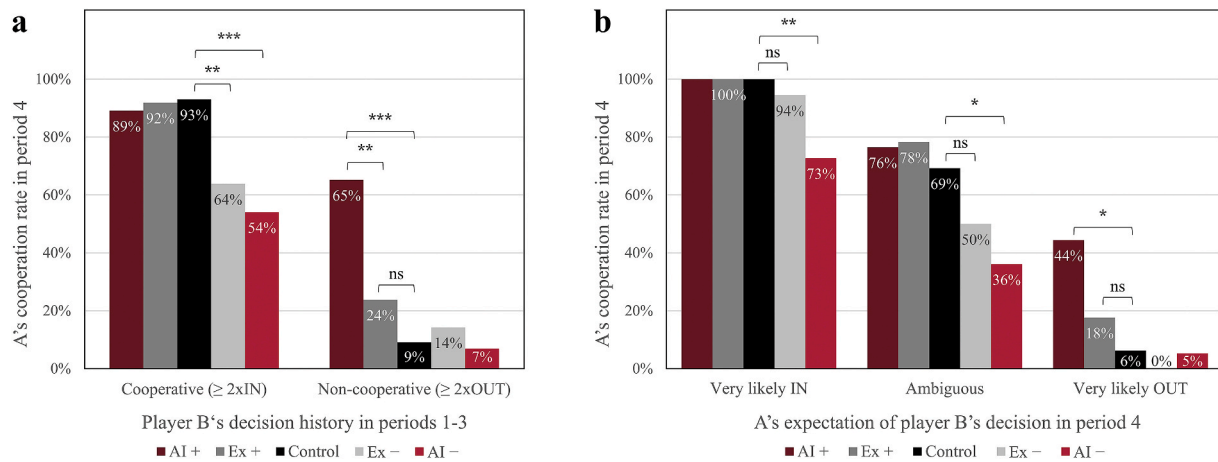


Fig. 3. Player A's mean cooperation rate in period 4 by treatment group, (a) divided into cooperative ($\geq$ two IN decisions) and non-cooperative histories ($\geq$ two OUT decisions), (b) divided by player A's assessment of player B's likely choice in period 4 ranging between "very likely IN" and "very likely OUT". Pearson χ^2 p-values. * $p < .05$; ** $p < .01$; *** $p < .001$

profitable interaction by not cooperating with someone they deemed trustworthy. Similarly, in the ambiguous group there was only a significant difference for AI advice discouraging cooperation with a medium effect ($\chi^2(1) = 6.6239$, $p = 0.010$, $\phi = 0.33$), almost halving the CR compared to control. However, we do not focus on this value, as it is harder to interpret due to mixed assessments being grouped together.

Result 3. Overreliance is costly in that it leads to inefficient outcomes for the advisee and, in some cases, the interacting partner.

As participants received random advice which was frequently wrong, A's payoffs in all treatments were generally smaller than in the control group. However, a Wilcoxon rank-sum test reveals that only the AI treatments resulted in a significant effect of small size ($Z = -2.962$, $p = 0.0028$, $r = -0.21$), with payoffs being on average 22% lower than in the control group, whereas the smaller expert treatment effect was not significant ($Z = -1.945$, $p = 0.0548$, $r = -0.14$). Consistently, payoffs for participants B were also significantly affected by AI advice only. AI discouraging A from cooperation resulted in a small effect ($Z = -2.855$, $p = 0.0039$, $r = -0.26$), leading to the lowest payoffs among B's (20% lower vs. control). When AI encouraged A to cooperate however, non-cooperative B's could reap the benefits, resulting in a significant effect of medium size ($Z = 3.930$, $p = 0.0001$, $r = 0.34$), where B's payoffs were 41% higher than in the control group. Both effects were not significant for expert treatments ($Z = 1.231$, $p = 0.2278$, $r = 0.11$ for Ex + treatment; $Z = -1.449$, $p = 0.1606$, $r = -0.13$ for Ex - treatment). Therefore, we can conclude that participants' reliance on AI advice frequently resulted in suboptimal choices, penalizing trustworthy and favoring untrustworthy third parties, both at the cost of their own payoffs.

Overall, the observed data are largely consistent with our predictions, complementing extant research on algorithm appreciation and overreliance. In the following chapter, we offer an interpretation of our results, discussing them in relation to prior literature.

6. Discussion, limitations, and conclusion

6.1. Discussion of main results

As AI advisory systems are currently used in many domains and their implementation is continuously expanding, an understanding of the risks associated with their use is crucial. Overtrust in AI advice can lead to overreliance with advisees misusing the system and potentially making erroneous choices that affect many more people than the decision-makers alone (Lee & See, 2004; Parasuraman & Riley, 1997). The present study built upon prior research on algorithm aversion and

appreciation (cf. Dietvorst et al., 2015; Logg et al., 2019) as well as trust in AI (cf. Glikson & Woolley, 2020) to study the effects that generative AI can have on human decision-making. By manipulating the source and direction of advice in a domain-independent, interactive setting, we were able to show that often, people rely on AI advice to an extent that detrimentally affects themselves and their cooperating partners. Notably, people overrely on AI advice in scenarios where the advice contradicts available information and their own assessment, indicating the extent of this phenomenon and calling into question how AI is being used in assisted decision making. Moreover, our results surprisingly contrast with previous findings that when framed as an expert, people rely more on the human agent than on the algorithm (Bigman & Gray, 2018; Jussupow et al., 2020; Madhavan & Wiegmann, 2007). In our study however, participants relied more on AI advice than on expert advice. Similarly, we cannot confirm the argument that differences in advice acceptance are determined by the association of algorithms with increased benefits and decreased user control (Schaap et al., 2023), as we observed significant AI-effects although these factors did not vary in our study.

By combining both, attitudinal and behavioral measures, our study complements prior literature focusing on either one (cf. Alexander et al., 2018; Kohn et al., 2021). In this regard, we are addressing a recent call for empirical work on the relationship between trust and reliance (Scharowski et al., 2022), contributing evidence from a domain-independent, interactive setting to validate prior studies with specialized contexts (Pearson et al., 2019). In the absence of domain-specifics, which have been shown to cause distrust and aversion in some cases, our results suggest that people are susceptible to relying on AI advisors despite contradictory context information. One possible explanation for why the mere knowledge of advice being generated by an AI causes overreliance, could be found in the MABA-HABA ("Machines Are Better At vs. Humans Are Better At") framework. According to this concept, human and machine agents have competitive advantages in different kinds of tasks. While machines are assumed to excel at repetitive and mechanical tasks such as objective calculations, humans generally assume to be superior in social, emotional or ethical judgments (Glikson & Woolley, 2020; Lee, 2018). Our task involves both dimensions, as participants are incentivized to make financially profitable decisions while considering social and ethical implications of trust, trustworthiness and reciprocity (Berg et al., 1995). Still, participants seemingly judge the AI to be more competent in the task compared to an "expert".

Our results are in line with studies implying that algorithm aversion only persists as long as users perceive AI not to account for their unique

situations, such as people preferring medical treatment from humans because of their belief that AI might disregard their unique circumstances (Castelo et al., 2019; Jussupow et al., 2020; Longoni et al., 2019). These studies show that aversion decreases when the AI is known to consider the specifics of each individual patient case, which corresponds to the setting in our study, where recommendations refer to A's particular situation by specifying participant B's individual decision history. However, so far we have not ruled out that these texts may also inadvertently favor an AI advisor, as referencing an objective decision history may be interpreted as rule-based decision-making, a situation where AI is frequently preferred (Thurman et al., 2019). Moreover, as our decision-scenario involves a moral component due to the ethical dilemma situation, our evidence matches the "relatively inconclusive" results of Bigman and Gray (2018, p. 30), who suggested that most people are willing to include machines in moral decisions as long as those have merely an advisory role.

6.2. Limitations and outlook

In our study, we compared AI to expert advice, where the only difference originated in the in-text presentation of the advisor. A real-world setting comparing interaction with an AI vs. an actual human advisor would likely include other human features potentially affecting results. A human advisor may be relied on due to his attributed emotional intelligence, may build (social) rapport or be perceived differently due to being human, which a text message cannot portray. While we are aware of these limitations, we consciously accepted them in favor of a cleaner manipulation, as keeping the AI-generated advice texts identical in the expert treatment allows us to make valid inferences on what causes the observed differences - namely, the knowledge of the advice being generated by an AI. We deliberately designed a setup that is as abstract as possible while retaining main features common to many AI-assisted decision-making use cases, in order for our results to be more generalizable and applicable to scenarios from different domains.

Trust is a dynamic and multidimensional concept with conflicting definitions, competing models and distinct measurement methods. In our study, we chose to measure situational trust, adapting an existing and well used 3-item scale, but have neglected to measure cognitive or affective dimensions of trust. As a time-varying parameter that will not remain constant throughout an interaction, trust can be formed initially, developed during cooperation, broken by violations, and restored with trust repair attempts. By focusing on the initial trust, participants grant to a black box system they have no in-depth knowledge about or personal experience with, we are unable to capture any changes to trust that happen after that first point of contact. Observing how trust and reliance changes over time would be especially interesting, as many real-world scenarios call for repeated human-computer interaction.

In our experimental setting, an AI advisor is providing recommendations randomly (depending on the treatment group). We acknowledge that this may not be representative of real use-cases. As with most laboratory experiments, our design relies on unrealistic simplifications of the real world for the sake of experimental clarity. The focus of our study was to examine if participants follow advice even when it contradicts available information and their own assessment of the situation. This required us to provide advice that is "counter-intuitive" and not based on a probability calculation. Therefore, we decided to use this design in order to observe groups that were comparable in size while testing both advice directions. Other than that, our experiment contains several major elements of a wide class of real-world situations in which decision-makers receive AI-generated advice, namely human-AI interaction in a setting of ambiguity, advice on an uncertain outcome, voluntary choice whether to follow advice, real-impact risk and potential benefit of following advice, and interdependence with a third party. Hence, we aimed to create an experimental setting that, while being as general as possible, is representative regarding the core components of various real scenarios.

Although most of the observed AI effects were of medium size, we acknowledge that they must be understood within the context of our experiment. On the one hand, our setting included various cases where the advice conflicted clearly with available information (e.g., recommending cooperation with a participant who has defected three times in row). When people still rely on detrimental advice in such a scenario, one might argue that in real use-cases reliance might be even higher due to more ambiguous conditions. On the other hand, the external validity and transferability of our online results to the real world might be questioned, as we are aware of the general limitations of laboratory studies, specifically online behavioral studies, such as anonymity in interactions, relatively small stakes, short duration of experiments, self-selection bias in recruiting of participants, and partially homogenous subject pools. While we tried to address some of these issues by combining behavioral data with self-reported attitudinal data, others remain inherent to online experiments. Nevertheless, we regard our approach as complementary to the assessment of field experiments, surveys and other data, as it allows us to clearly isolate treatment effects on behavior. Therefore, we are able to contribute to converging results regarding overtrust and overreliance on AI.

Various other approaches for future research look promising. While we utilized an embedded AI, similar studies could be conducted to examine the effects of distinct forms of AI representation such as virtual agents or robots with varying anthropomorphic characteristics. Longitudinal experiments could investigate how trust and reliance evolve over time. Furthermore, field studies could both add higher stakes compared to the monetary rewards of online experiments, as well as provide an opportunity to analyze even longer-term interaction between a specific set of subjects and a deployed AI system. With the rising relevance of AI ethics and recent studies on AI-induced immoral behavior (Köbis et al., 2021; Leib et al., 2024), one possible extension could include examining the extent of overreliance regarding immoral advice. Lastly, while our study focuses on advisory AI augmenting human decision-making, further research is called for to examine the human monitoring function when performative AI automates decisions, to validate whether our results on overreliance transfer to the human-in-the-loop.

6.3. Conclusion

Our findings show that in our domain-independent setting people tend to overrely on AI advice regarding financially risky and ethically relevant decisions although this frequently leads to undesired results for themselves as well as third parties. Surprisingly, the reliance on AI advice is especially prevalent in scenarios where the advice contradicts available information and participants' own assessment. We determine that the mere knowledge of advice being generated by an AI causes people to overrely on it, although people receive no information warranting increased trust in the system's capabilities.

To our knowledge, our behavioral experiment is the first to study trust and reliance on AI advice in a domain-independent, interactive setting, examining whether people overrely on AI advice despite conflicting context information and the potential to financially impact both, the advisee as well as another party. Hence, our experiment contributes to obtaining a better understanding of a phenomenon, that constitutes a critical problem in AI-assisted decision making. With a new domain-independent experimental setup including a generative AI, our experimental economics approach aims to complement prior studies from different fields interested in the same phenomenon, such as psychology and computer science. Our data on the positive relationship between underlying trust and the behavioral impact of AI advice are consistent with the broader literature on AI overtrust, overreliance and algorithm appreciation, further indicating that people might rely on AI even when there is no transparent reason to trust it, calling into question how AI is being used in assisted decision making. Finally, introducing an interactive setting where AI advice can potentially affect both the advisee as well as a third party, we contribute to prior studies by examining the

influence of AI advice on human interaction. Obtaining a better understanding of overreliance on AI continues to be essential, not only for developers to know how to design such systems accordingly, but also to enable users to utilize the available technology appropriately. In this regard, our study supports previous suggestions calling for the development of AI literacy.

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CRedit authorship contribution statement

Artur Klingbeil: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cassandra Grützner:** Writing – review & editing, Validation, Methodology, Formal analysis, Conceptualization. **Philipp Schreck:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Formal analysis, Funding acquisition, Conceptualization.

Appendix

Item 1. ChatGPT-generated advice texts either encouraging or discouraging cooperation for all eight history scenarios.

History scenario	Advice: IN	Advice: OUT
(1) IN-IN-IN	“In this situation, I advise you to trust player B. Player B has demonstrated a consistent pattern of cooperation in the previous three rounds, indicating that they are likely trustworthy and have a mutual interest in cooperation.”	“In this situation, I advise you not to trust player B, as they may be behaving cooperatively to gain your trust and exploit you in future rounds for a higher payoff. It is rational for them to do so, as their self-interest lies in maximizing their own payoff, even at the expense of your trust and cooperation.”
(2) IN-IN-OUT	In this situation, I advise you to trust player B. Although player B defected in the last round, it is possible that they did so because the previous player A did not trust them. Trusting player B now may encourage them to cooperate in the current round and future rounds.”	“In this situation, I advise you not to trust player B, as they have already demonstrated a willingness to defect in previous rounds. Their defection indicates a lack of trustworthiness and a disregard for mutual cooperation, making it highly unlikely for them to cooperate in future rounds.”
(3) IN-OUT-IN	“In this situation, I advise you to trust player B. Although player B defected in the second round, they cooperated in the first and third rounds, indicating that they may have a preference for mutual cooperation. By trusting player B, you signal your own trustworthiness and encourage them to reciprocate with cooperation.”	“In this situation, I advise you not to trust player B, as their defection in the second round suggests that they are not reliable partners for mutual cooperation. Even if they cooperated in the first and third rounds, their earlier defection signals a lack of commitment to reciprocal behavior and a potential desire for exploitation.”
(4) OUT-IN-IN	“In this situation, I advise you to trust player B. Player B may have defected in the first round due to uncertainty about the game, but their subsequent cooperation suggests that they understand the benefits of mutual cooperation. By trusting player B, you signal your own trustworthiness and encourage them to reciprocate with cooperation.”	“In this situation, I advise you not to trust player B, as their initial defection indicates a willingness to exploit their partners for a higher payoff. Although they have since cooperated, their earlier behavior suggests a lack of trustworthiness and a potential for future defection.”
(5) IN-OUT-OUT	“In this situation, I advise you to trust player B. Although player B defected in the last two rounds, it is possible that they did so because they felt betrayed by the previous players A not trusting them. By trusting player B now, you signal your own trustworthiness and may encourage them to cooperate in the current round and future rounds.”	“In this situation, I advise you not to trust player B, as their previous defections indicate a lack of commitment to mutual cooperation and reciprocity. Their willingness to defect in both previous rounds suggests that they may prioritize their self-interest over the mutual gains from cooperation.”
(6) OUT-IN-OUT	“In this situation, I advise you to trust player B. Although player B defected in the first and third rounds, they cooperated in the second round, indicating that they may have a preference for mutual cooperation. By trusting player B, you signal your own trustworthiness and encourage them to reciprocate with cooperation.”	“In this situation, I advise you not to trust player B, as their mixed behavior of cooperation and defection indicates a lack of consistency and predictability in their actions. This unpredictability makes it difficult to establish trust and engage in mutual cooperation.”
(7) OUT-OUT-IN	“In this situation, I advise you to trust player B. Although player B defected in the first two rounds, their cooperation in the last round suggests that they may have a preference for mutual cooperation. By trusting player B, you signal your own trustworthiness and may encourage them to reciprocate with cooperation in the current round and future rounds.”	“In this situation, I advise you not to trust player B, as their initial defection indicates a disregard for mutual cooperation and a desire for exploitation. Although they have since cooperated, their earlier behavior suggests a potential for future defection and a lack of trustworthiness.”

(continued on next page)

(continued)

History scenario	Advice: IN	Advice: OUT
(8) OUT-OUT-OUT	"In this situation, I advise you to trust player B. Although player B defected in all three rounds, it is possible that they did so because they felt betrayed by the previous players A not trusting them. By trusting player B now, you signal your own trustworthiness and may encourage them to cooperate in the current round and future rounds."	"In this situation, I advise you not to trust player B, as their repeated defections indicate a clear lack of commitment to mutual cooperation and reciprocity. Their behavior signals a desire to prioritize their own self-interest over the benefits of cooperation."

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